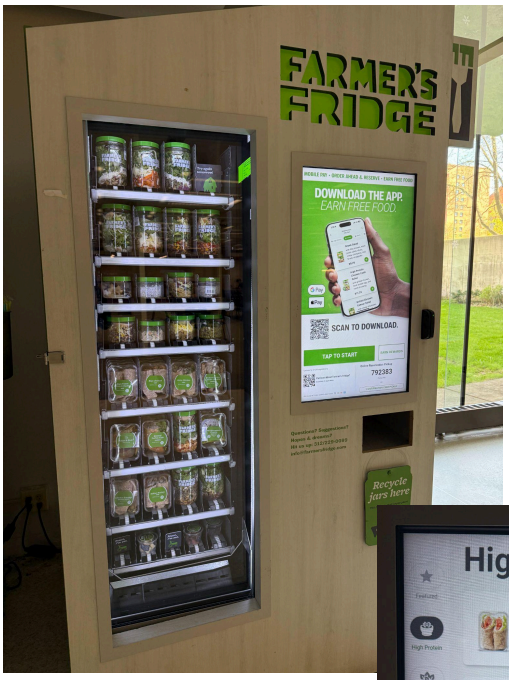
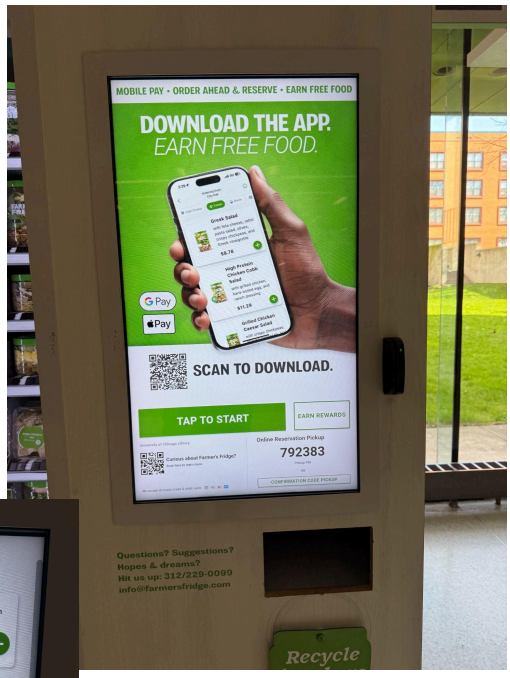
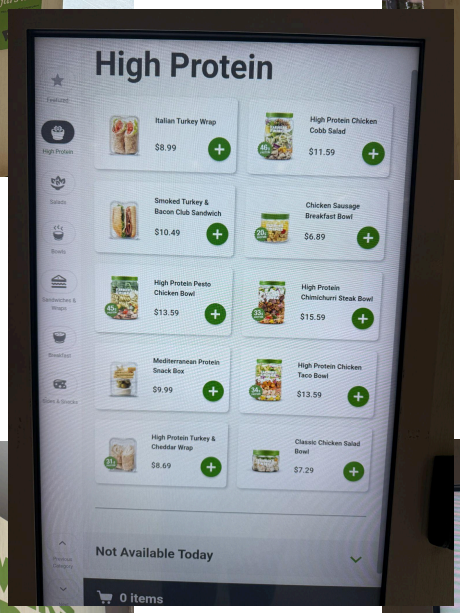
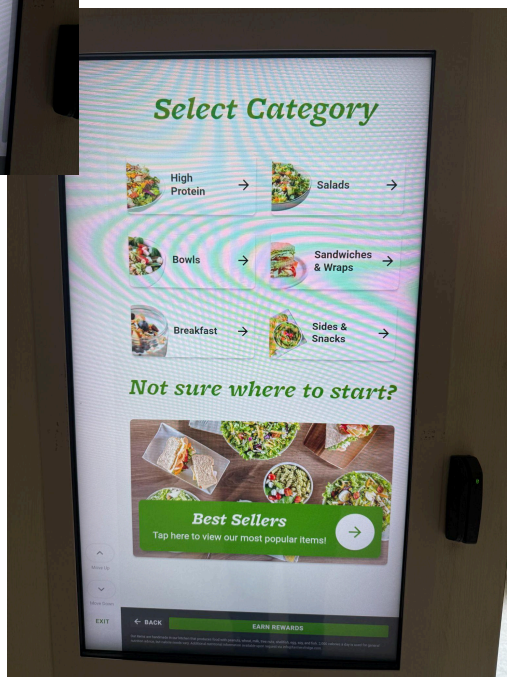
For this assignment I looked at the Farmer’s Fridge vending machine.









What kinds of problems did people have using the system?

One issue I observed was not necessarily a problem with the system itself, but with how people interacted with it. Many people looked through the glass side of the vending machine to decide whether they wanted anything, rather than browsing the menu on the screen. It seemed that viewing the items from the side was less useful than navigating the UI, but some users seem to have preferred this faster option rather than spending extra time scrolling through the menu. This behavior may be linked to the interface design as the menu was divided into categories without an “all items” page which made it somewhat tedious to view the full selection.

Another problem users encountered was the unavailability of utensils after they had already completed their purchase. I was unsure how they ended up eating their meals since they simply walked away after noticing. This was a design problem as the system does not prompt users to check whether utensils are available before ordering or communicate utensil availability clearly on the screen. This could be considered a minor usability issue or perhaps a dark pattern as users are not informed of an important constraint before committing to their purchase.

What aspects of the system appeared to be easy to learn?

The ordering process was generally intuitive and closely resembled other ordering systems like McDonalds'. Users simply tapped to start, selected a food category, added items to their order, and proceeded to payment. This flow made it easy for both experienced and inexperienced users to quickly understand how to navigate the system. Compared to a nearby boba machine which made users tap their card before selecting an item, this system followed a more conventional sequence where users would pay at checkout.

Additionally, most users appeared to understand where to find utensils even though they were located on the side of the machine and were not physically visible from the front view. However, there is a utensil sign attached to the right of the vending machine which may have acted as a sufficient indicator. With every user consistently going towards the utensil station, this suggests that this design was intuitive.

What aspects of the system appeared to be difficult to learn?

While the system was generally easy to learn, there were a few minor usability challenges. For example, I initially assumed that the machine would support tap-to-pay using a phone, but it only accepted card payments via swipe. This limitation was not immediately obvious and could lead to frustration especially for users who do not carry their wallets. Although this is quickly discovered during the process, it still results in wasted time if users select items before realizing they cannot complete the transaction.

Another challenge was the lack of detailed information about the items. The interface was simple, but this simplicity meant details such as nutritional information, including calories or macronutrients only appeared within the selected item's individual page. This is particularly important given that some items are marketed as “high protein” or as snacks, where such

information could influence user decisions and getting to see these details among all the options can allow users to make comparisons.

Users who selected items by looking through the glass sometimes struggled to locate those items within the digital menu. Because the menu was divided into multiple categories and not all items in the machine were clearly labeled with their category, users often had to scroll through several categories to find what they were looking for.

How might the design of the system be improved?

One improvement would be to include calorie counts or basic nutritional information beneath each item so that users can easily make comparisons. There is sufficient unused space on the interface, and adding this information could help users make decisions quickly.

Additionally, the system would benefit from an “all items” view that allows users to browse the entire selection without navigating through multiple categories. To further reduce friction, the UI could be designed to mirror the physical layout of the vending machine by aligning digital selections with the actual row and column positions of items. This would make it easier for users who rely on visual selection through the glass.

Another improvement would be to ensure that all items are clearly labeled with names, as some currently rely too heavily on images alone. This can lead to confusion if the image is unclear or does not fully represent the product. Having all items labeled would also allow the UI to implement a search bar so users can easily find their items if they already had a selection in mind.

The system should also notify users if essential items, such as utensils, are out of stock before they complete their purchase. While this change might slightly reduce sales, it would significantly improve transparency and overall user experience.

Tell us the process you followed in the report to use your agent to create the new system design

I first planned out the features I wanted to add and documented them in a Markdown file as my prompt. I then asked Kiro to summarize my ideas and clarify any uncertainties. Based on this, Kiro generated a requirements Markdown file and divided the work into tasks. I have included the original Markdown file I used as my prompt in `plan.md`. In addition to this prompt, I also provided images of the vending machine UI to give Kiro a clearer understanding of what it should recreate and what it should improve. Once Kiro was done with the tasks. I then went through the html by simulating both website and mobile UI to look for bugs. Many of the issues I encountered were related to alignment of the UI such as improperly centered elements, and text

or emojis breaking beyond their container boundaries. I then provided Kiro with a list of fixes I wanted it to complete until I was satisfied with the result.

Give an estimate of how long it took you to code the interface in your report
Kiro ran the code to build the website for about an hour and a half. I then spent about thirty minutes bug fixing the UI.